Operations Research 1: Deterministic Models
Linear Programming: Introduction and Modeling

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Fall, 2025

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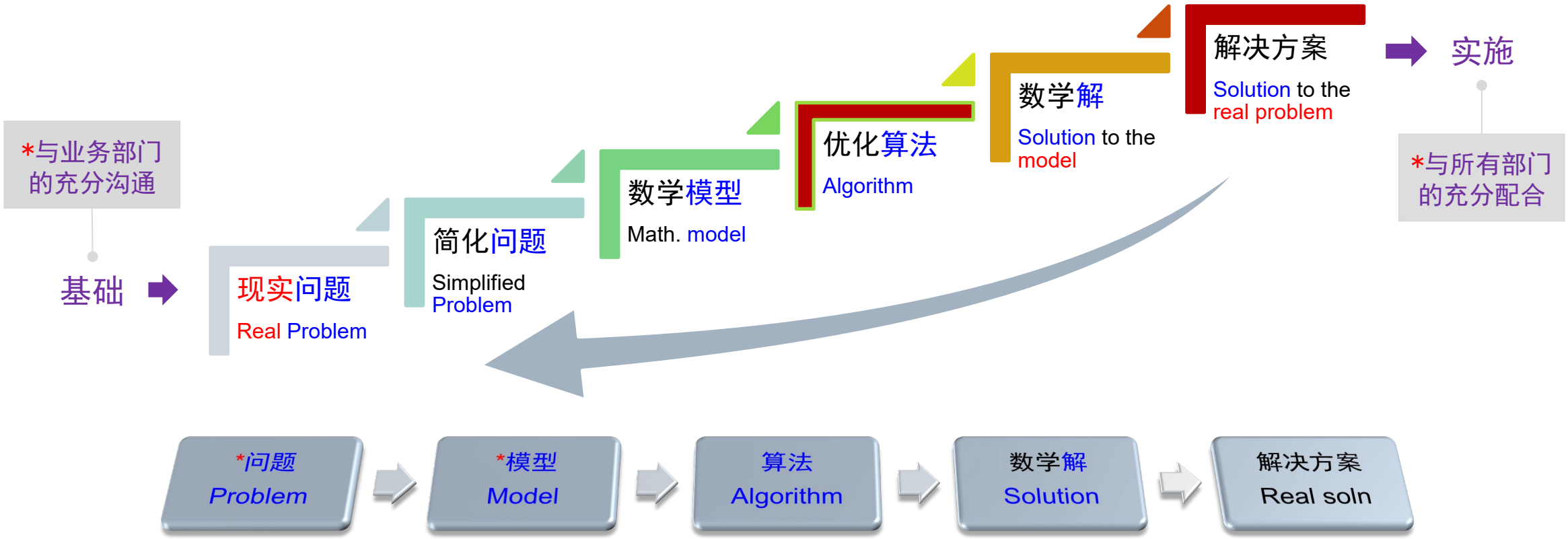
Department of Industrial Engineering
Tsinghua University, China

运筹学应用路径
OR Application Roadmap

OR Application Roadmap



Source: TOpS Lab (2020)

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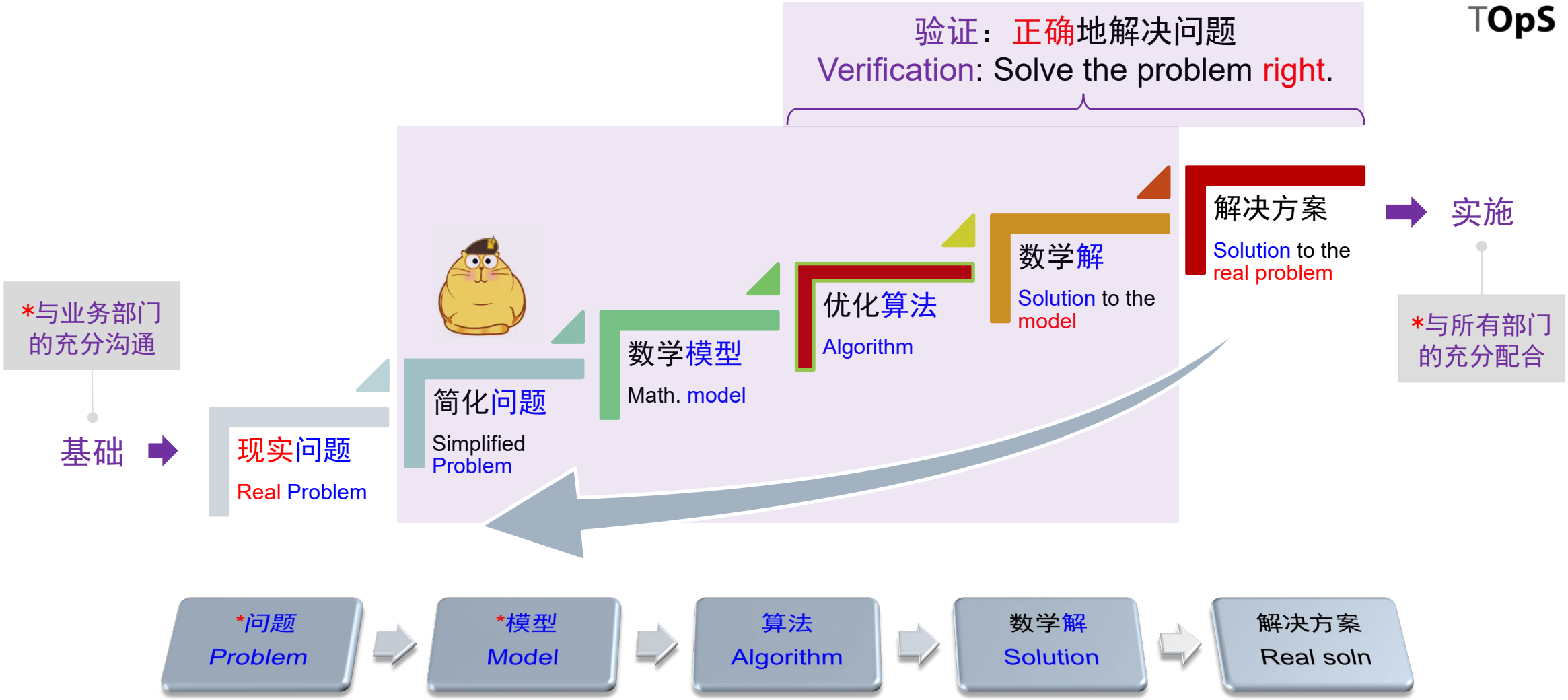

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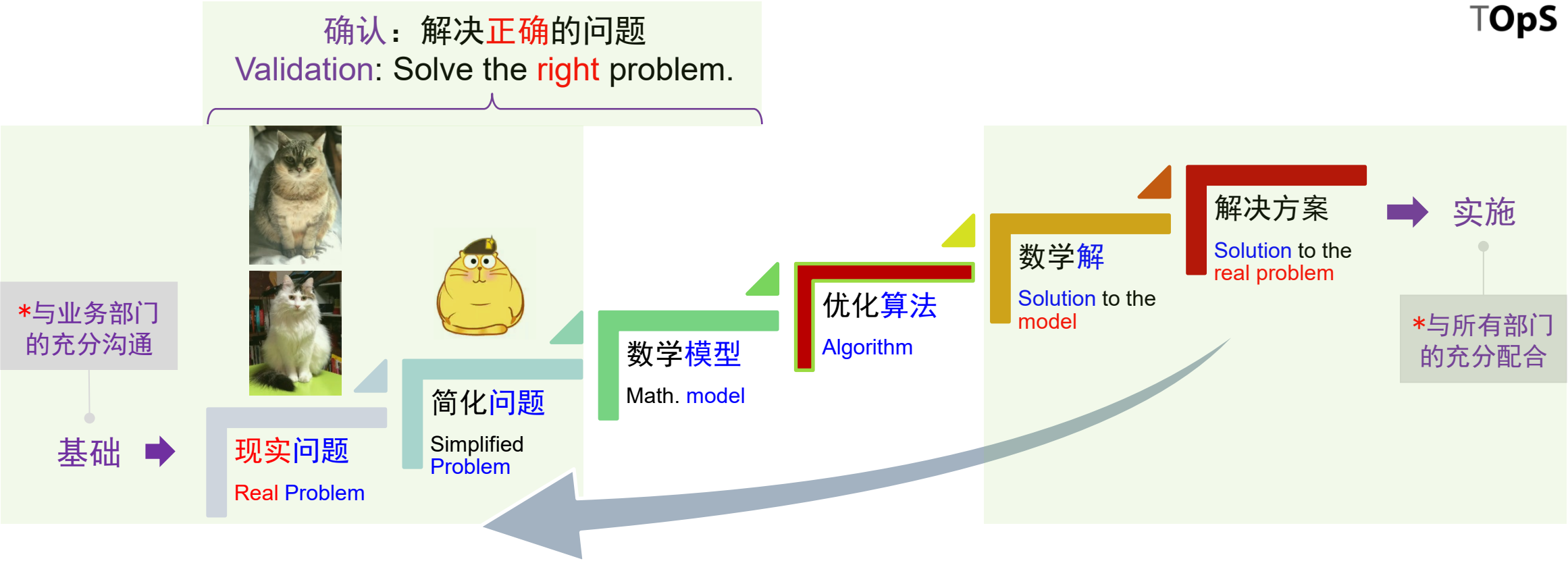

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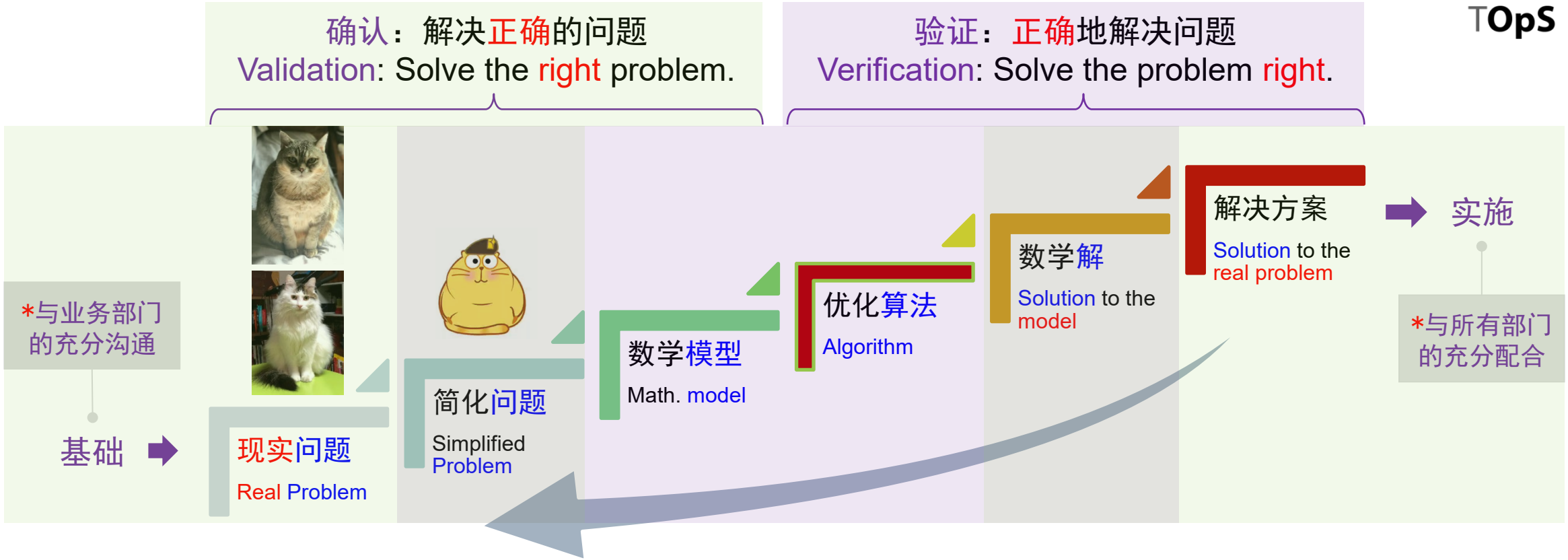

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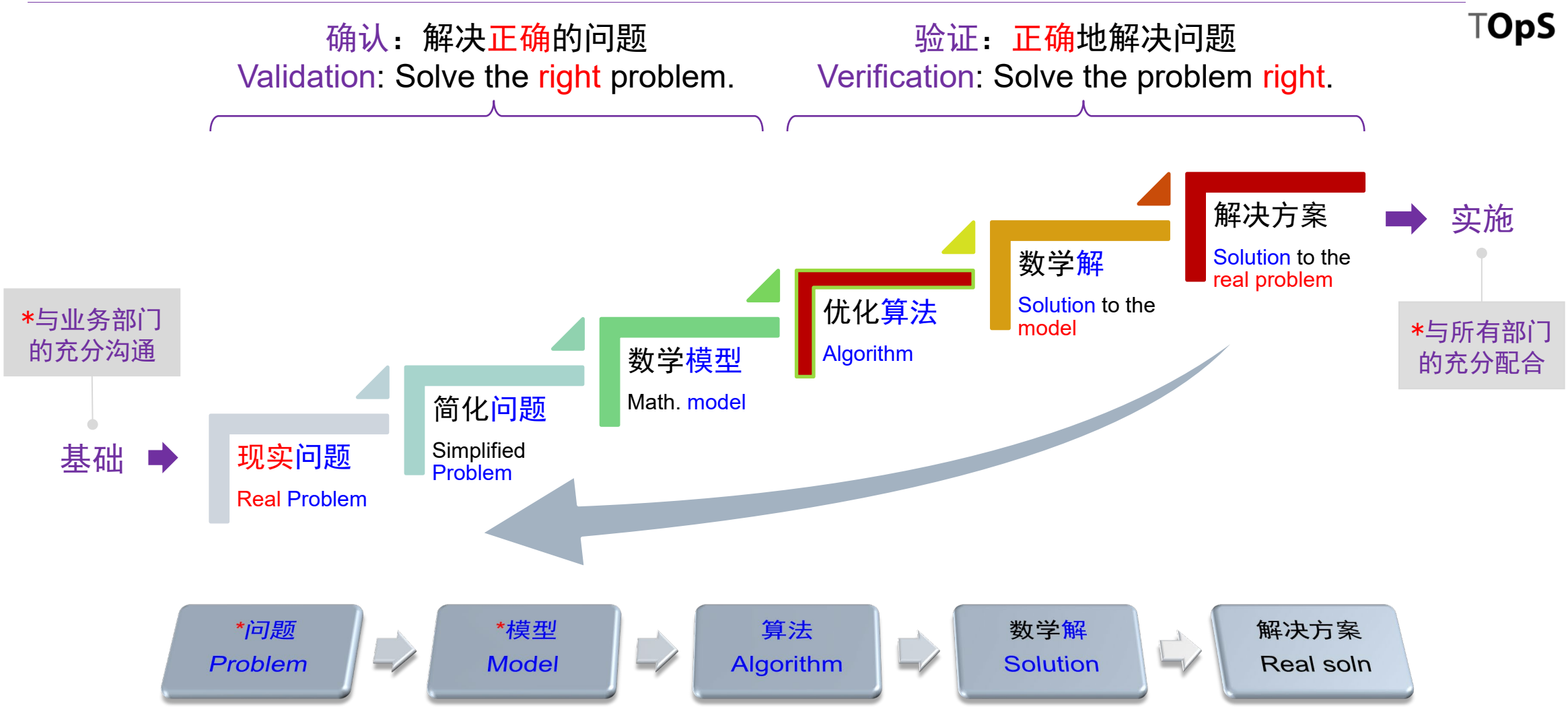

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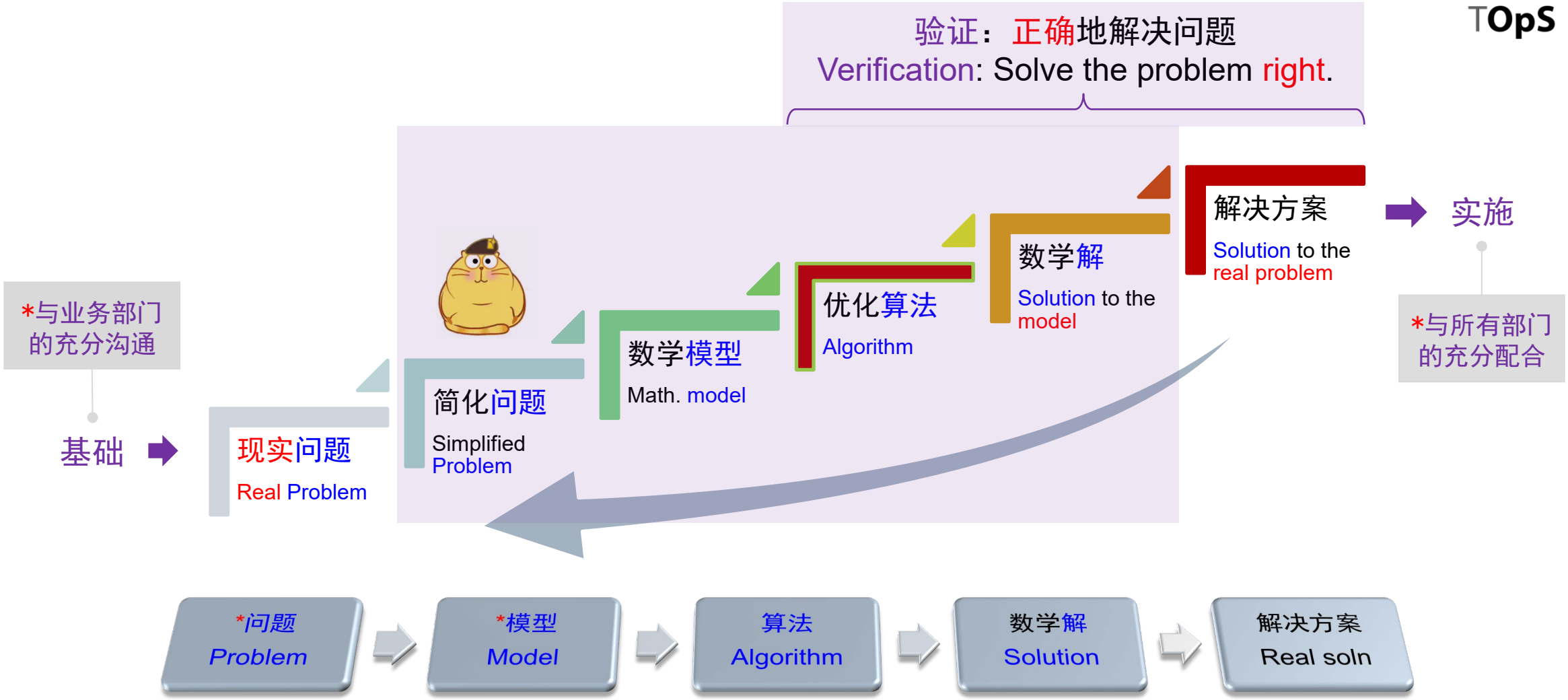


Source: TOpS Lab (2020)

弹 Discussions: What Is a Decision?

#Discussions: What Is a Decision?

- (Informally) A decision is a choice.
- When you make a choice, there are alternatives.
- Some alternatives are possible, whereas some are not.
  - Constraints ➔ feasibility
- When you choose one from the alternatives, you need a way to evaluate the alternatives, *explicit* or *implicit* (or even *subconscious*), *quantitative* or *qualitative*.
  - Objective "function" ➔ optimality
- You make your choice in certain context.
  - The decision problem ➔ context dependent

弹

*Optimization Problem/Model: Key Elements

- 我们知道什么? What do we know?

  流程 + 数据 process + data

- 我们需要决定什么？ What do we need to decide?

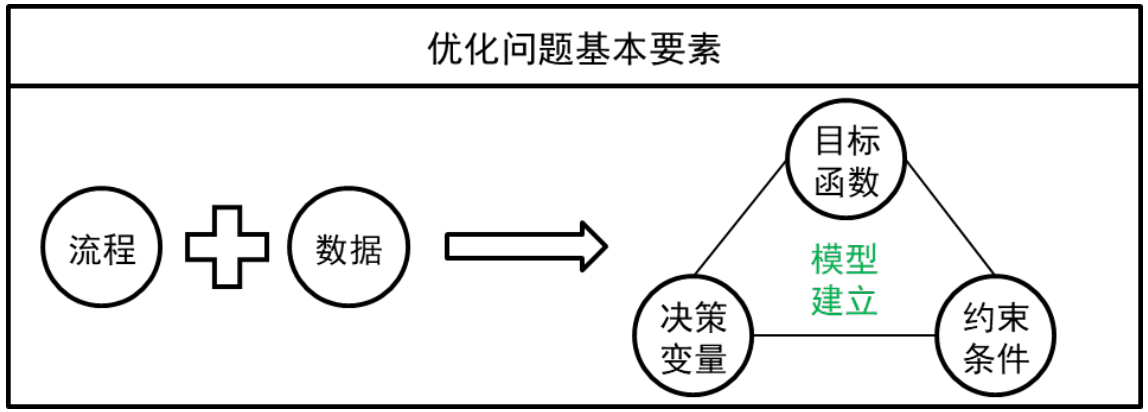
  决策变量 decision variables

- 我们希望达到什么目标？ What do we want to achieve?

  目标函数 objective function

- 我们的决策受到什么限制？ What prohibits our ability to do so?

  约束条件 constraints



Resource Allocation Problems

Resource Allocation Problems

- Allocating limited resources among competing activities in a best possible (i.e. *optimal*) way.

Source: Hillier and Lieberman (2021)

*Age of Empires II

- Four ages
  - The Dark Age
  - The Feudal Age
  - The Castle Age
  - The imperial Age
- 30+ Civilizations

- Starting a game
  - Town center
  - Villagers
  - A scout cavalry
  - Starting resources

Source: https://www.ageofempires.com

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Resource Allocation Problems

- Allocating limited resources among competing activities in a best possible (i.e. *optimal*) way.

- Activities ➔ levels of activities ➔ consumption of resources
  - Decision variables
- Limited resources ➔ necessary to perform activities
  - Constraints
- Overall measure of performance
  - Objective function

Source: Hillier and Lieberman (2021)

Linear Programming

Linear Programming

- Linear ➔ linear functions
  - Constraints & objective functions
  - With regard to decision variables

- Programming ➔ “planning”

Example: Product Mix (Wyndor Glass Co.)

Wyndor Glass Co.

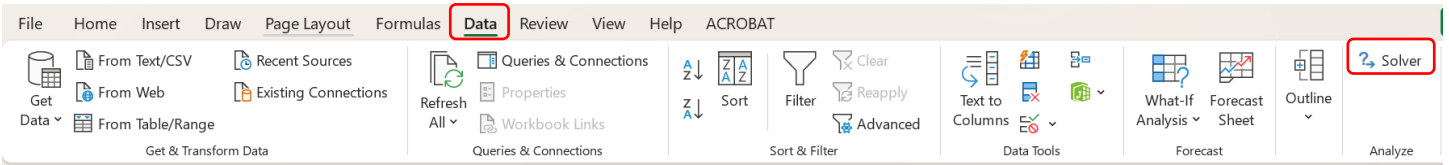
- Two products
  - Product 1
    - Glass door w/ aluminum framing
  - Product 2
    - Wood-frame window
- Three plants
  - Plant 1
    - Aluminum frames and hardware
  - Plant 2
    - Wood frames
  - Plant 3
    - Glass and product assemble
- Data
  - Available production time in each plant (hours / week)
  - Production time used in each plant for each batch of each product (hours / batch)
  - Profit per batch of each product

	Doors	Windows		
Profit per batch	$3,000	$5,000		
				Hours
	Hours used per batch produced			available
Plant 1	1	0		4
Plant 2	0	2		12
Plant 3	3	2		18

- What do we need to decide?
- How do we evaluate our decision?

*Wyndor Glass Co.

- Excel Solver
  - Data ➔ Solver
  - Installation: File ➔ Options ➔ Add-Ins ➔ Excel Add-ins ➔ Solver Add-in



- Excel: Range names
  - Formulas ➔ Name Manager
  - Shortcut: Ctrl + Shft + F3 (& F3 to show the defined range names)
    - Mac: Press and hold the fn key to display the F keys (F1-F12) in the touch bar.

Wyndor Glass Co. Product-Mix Problem							Range names	
							Batches_produced	=Solution!$B$12:$C$12
	Doors	Windows					Hours_available	=Solution!$F$7:$F$9
Profit per batch	$3,000	$5,000					Hours_used	=Solution!$D$7:$D$9
			Hours		Hours		Hours_used_per_batch_produced	=Solution!$B$7:$C$9
	Hours used per batch produced		used		available		Profit_per_batch	=Solution!$B$4:$C$4
Plant 1	1	0	2	<=	4		Total_profit	=Solution!$F$12
Plant 2	0	2	12	<=	12			
Plant 3	3	2	18	<=	18			
	Doors	Windows			Total profit			
Batches produced	2	6			$36,000			

d

*Wyndor Glass Co.

Excel Solver

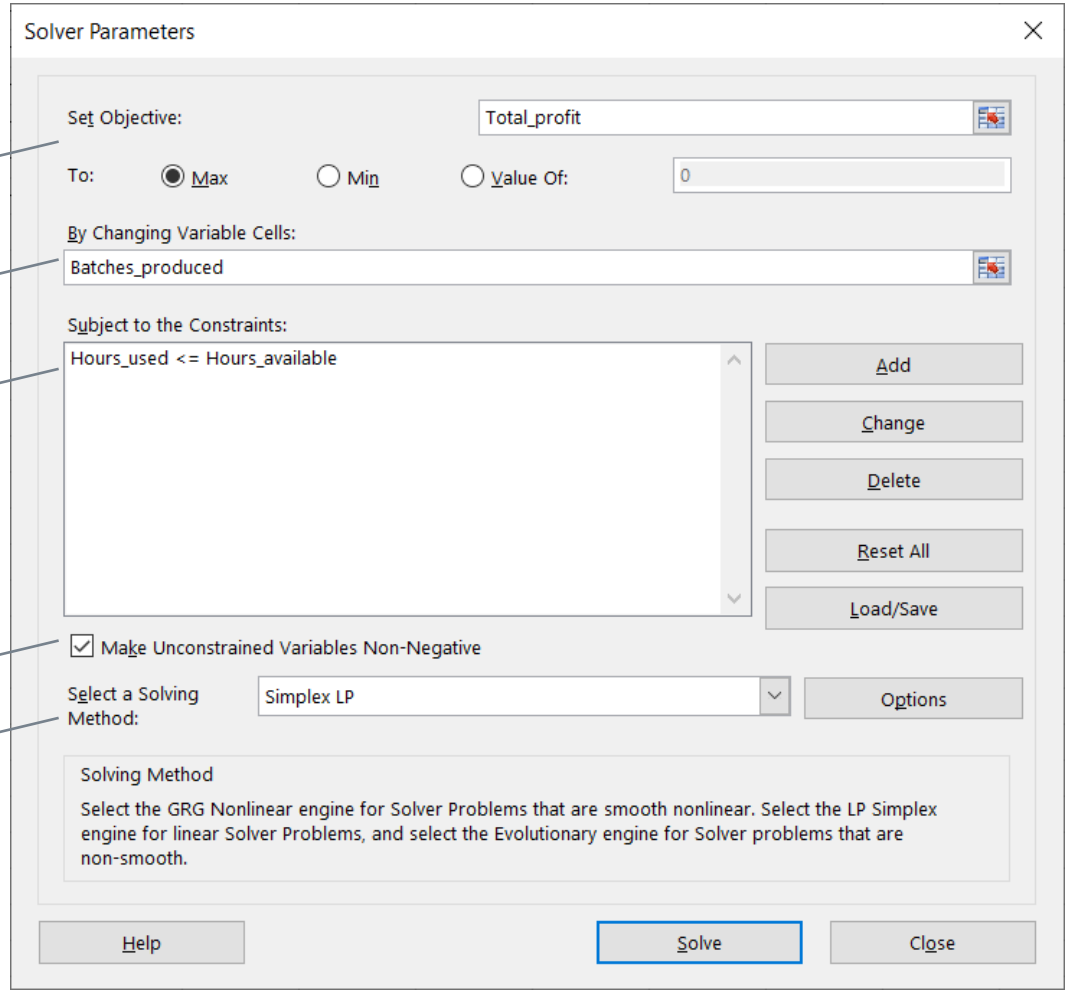
Objective function

Decision variables

Constraints

Nonnegativity

Solution method



Wyndor Glass Co.

- Mathematical formulation
  - x_1 = # batches of product 1 produced per week
  - x_2 = # batches of product 2 produced per week
  - Z = total profit per week

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	Doors	Windows			Total profit
Batches produced					

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\begin{aligned}
\max \quad & Z = 3x_1 + 5x_2 \\
\text{s.t.} \quad & x_1 \le 4, \\
& 2x_2 \le 12, \\
& 3x_1 + 2x_2 \le 18, \\
& x_1, \quad x_2 \ge 0.
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Lingo Model - 4.1_wyndor_lingo

```
! The Wyndor Model

! x1 = doors, x2 = windows.
! objective;
max = 3*x1 + 5*x2;

! constraints @ plants 1, 2, & 3;
        x1          < 4;
            2*x2    < 12;
      3*x1 + 2*x2   < 18;
end
```

<=

*Standard Form of LP Models

Maximize $Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n$

Subject to

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &\le b_1, \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &\le b_2, \\
&\vdots \\
a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &\le b_m,
\end{aligned}
$$

and

$$x_1 \ge 0, \quad x_2 \ge 0, \quad \cdots, \quad x_n \ge 0.$$

$$
\begin{aligned}
\max \quad & Z = 3x_1 + 5x_2 \\
\text{s.t.} \quad & x_1 \le 4, \\
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≥ 0

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and

$$x_1 \ge 0, \quad x_2 \ge 0, \quad \cdots, \quad x_n \ge 0.$$

≥ 0

Maximize $Z = c^{\mathrm{T}}x$

Subject to

$$Ax \le b,$$
$$x \ge 0.$$

≥ 0

Standard Form of LP Models

Maximize $Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n$

Subject to

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Four Assumptions of LP

- Proportionality
  - Objective value/LHS is proportional to the level of activity
  - Nonlinear/mixed integer programming
- Additivity
  - Sum of individual contributions (objective/LHS)
  - Nonlinear programming
- Divisibility
  - (Mixed) Integer programming
- Certainty
  - Sensitivity analysis, but …
  - Stochastic programming, (approximate) dynamic programming, simulation optimization, robust optimization, …

Mathematical Programming Solvers

- Excel
- Lindo/Lingo
- IBM ILOG Cplex
- FICO Xpress
- **Gurobi**
- SAS
- AIMMS
- GAMS
- Frontline

$$
\begin{array}{lrl}
\max & Z = 3x_1 + 5x_2 & \\
\text{s.t.} & x_1 & \leq 4, \\
 & 2x_2 & \leq 12, \\
 & 3x_1 + 2x_2 & \leq 18, \\
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\end{array}
$$

- Free/open source solvers

Source: http://www.lindo.com/

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      3*x1 + 2*x2 < 18;
end
```

Solution Report - 2.1_wyndor_lingo

```
LINGO/WIN64 19.0.24 (26 Oct 2020), LINDO API 13.0.4099.232

  Licensee info: Eval Use Only
  License expires: 16 MAR 2025

  Global optimal solution found.
  Objective value:                              36.00000
  Infeasibilities:                              0.000000
  Total solver iterations:                             1
  Elapsed runtime seconds:                          0.07

  Model Class:                                        LP

  Total variables:                     2
  Nonlinear variables:                 0
  Integer variables:                   0

  Total constraints:                   4
  Nonlinear constraints:               0

  Total nonzeros:                      6
  Nonlinear nonzeros:                  0


                              Variable           Value        Reduced Cost
                                    X1        2.000000            0.000000
                                    X2        6.000000            0.000000

                                   Row    Slack or Surplus      Dual Price
                                     1        36.00000            1.000000
                                     2        2.000000            0.000000
                                     3        0.000000            1.500000
                                     4        0.000000            1.000000
```

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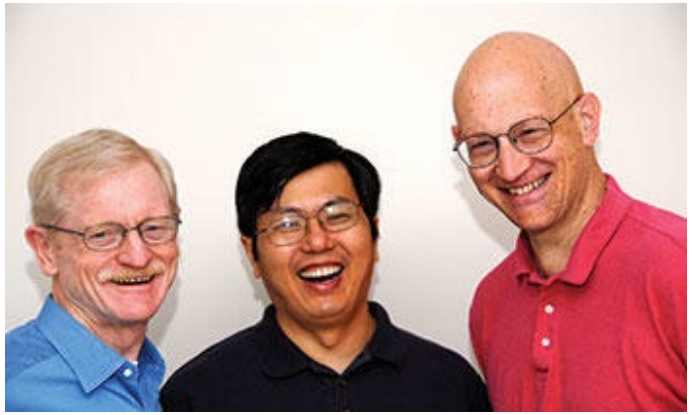

Bob **Bi**xby, Zonghao **Gu**, and Ed **Ro**thberg

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wyndor.py

```python
from wyndor_pandas import generate_data
from wyndor_gurobi import formulate_model, solve_model, write_model
#from wyndor_gurobi import formulate_model, read_model, solve_model, write_model

# Generate Pandas DataFrames directly.
products, products_plants, plants = generate_data()

# Formulate the LP instance
# Option 1: formulate manually
model = formulate_model(products, products_plants, plants)
# Option 2: read from a .lp or .mps file
# model = read_model("wyndor_read.lp")
# model = read_model("wyndor_read.mps")

# Write the LP instance as a .lp or .mps file.
write_model(model, "wyndor_write.lp")
# write_model(model, "wyndor_write.mps")

# Solve the LP instance.
solve_model(model)
```

```
Optimize a model with 3 rows, 2 columns and 4 nonzeros
Model fingerprint: 0xee02f900
Variable types: 0 continuous, 2 integer (0 binary)
Coefficient statistics:
  Matrix range     [1e+00, 3e+00]
  Objective range  [3e+03, 5e+03]
  Bounds range     [0e+00, 0e+00]
  RHS range        [4e+00, 2e+01]
Found heuristic solution: objective 27000.000000
Presolve removed 3 rows and 2 columns
Presolve time: 0.00s
Presolve: All rows and columns removed

Explored 0 nodes (0 simplex iterations) in 0.01 seconds (0.00 work units)
Thread count was 1 (of 8 available processors)

Solution count 2: 36000 27000

Optimal solution found (tolerance 1.00e-04)
Best objective 3.600000000000e+04, best bound 3.600000000000e+04, gap 0.0000%

The optimal solutions:
Batch_produced[Doors]: 2.0
Batch_produced[Windows]: 6.0
The optimal objective function value is:36000.0
```

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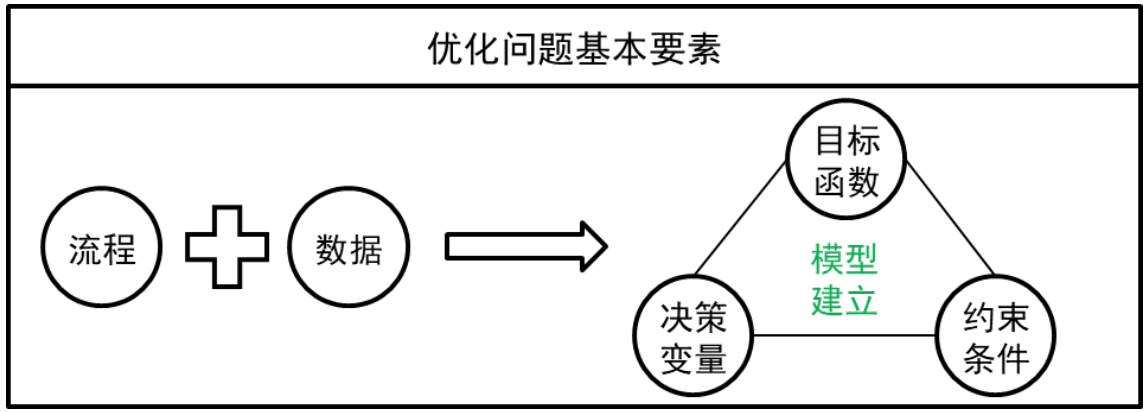

More Modeling Examples

Example: Regional Planning

Regional Planning

- Three communities: {1, 2, 3}
- Three crops: {sugar beets, cotton, sorghum}
- Resources/constraints
  - Usable land (community)
  - Water allocation (community)
  - Land quota (crop)
  - Unit water consumption (crop)
  - "Fairness"

Community	Usable land	Water
1	400	600
2	600	800
3	300	375

Crop	Land quota	Water consumption	Net return
1	600	3	$1,000
2	500	2	$750
3	325	1	$250

- Decisions
  - Acres allocated to each crop at each community
- Objective
  - Maximize the total net return ($) to the region

Regional Planning

- Data

Community	Usable land	Water
1	400	600
2	600	800
3	300	375

- Decisions

Crop	Land quota	Water consumption	Net return
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3	325	1	$250

- Model

*Regional Planning

- Data

$$L_j^U = \text{useable land of community } j \in J = \{1,2,3\}$$
$$W_j^A = \text{water resource of community } j \in J$$
$$L_i^Q = \text{land quota of crop } i \in I = \{1,2,3\}$$
$$W_i^C = \text{water consumption (per acre) of crop } i \in I$$
$$R_i = \text{net return (per acre) of crop } i \in I$$

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- Decisions

$$x_{ij} = \text{acres allocated to crop } i \in I \text{ at community } j \in J$$

- Model

*Regional Planning

- **Data**

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- **Decisions**

$$x_{ij} = \text{acres allocated to crop } i \in I \text{ at community } j \in J$$

- **Model**

$$\begin{aligned}
\max \quad & \sum_{i\in I} R_i \sum_{j\in J} x_{ij} && \text{maximize total net return} \\
\text{s.t.} \quad & \sum_{i\in I} x_{ij} \le L_j^U, \quad \forall j \in J, && \text{land constraints} \\
& \sum_{i\in I} W_i^C x_{ij} \le W_j^A, \quad \forall j \in J, && \text{water constraints} \\
& \sum_{j} x_{ij} \le L_i^Q, \quad \forall i \in I, && \text{land quota constraints} \\
& \frac{\sum_{i\in I} x_{i1}}{L_1^U} = \frac{\sum_{i\in I} x_{i2}}{L_2^U} = \frac{\sum_{i\in I} x_{i3}}{L_3^U}, && \text{fairness} \\
& x_{ij} \ge 0, \quad \forall i \in I, j \in J.
\end{aligned}$$

*Regional Planning

- **Data**

$$L_j^U = \text{useable land of community } j \in J = \{1,2,3\}$$
$$W_j^A = \text{water resource of community } j \in J$$
$$L_i^Q = \text{land quota of crop } i \in I = \{1,2,3\}$$
$$W_i^C = \text{water consumption (per acre) of crop } i \in I$$
$$R_i = \text{net return (per acre) of crop } i \in I$$

Community	Usable land	Water
1	400	600
2	600	800
3	300	375

Crop	Land quota	Water consumption	Net return
1	600	3	$1,000
2	500	2	$750
3	325	1	$250

- **Decisions**

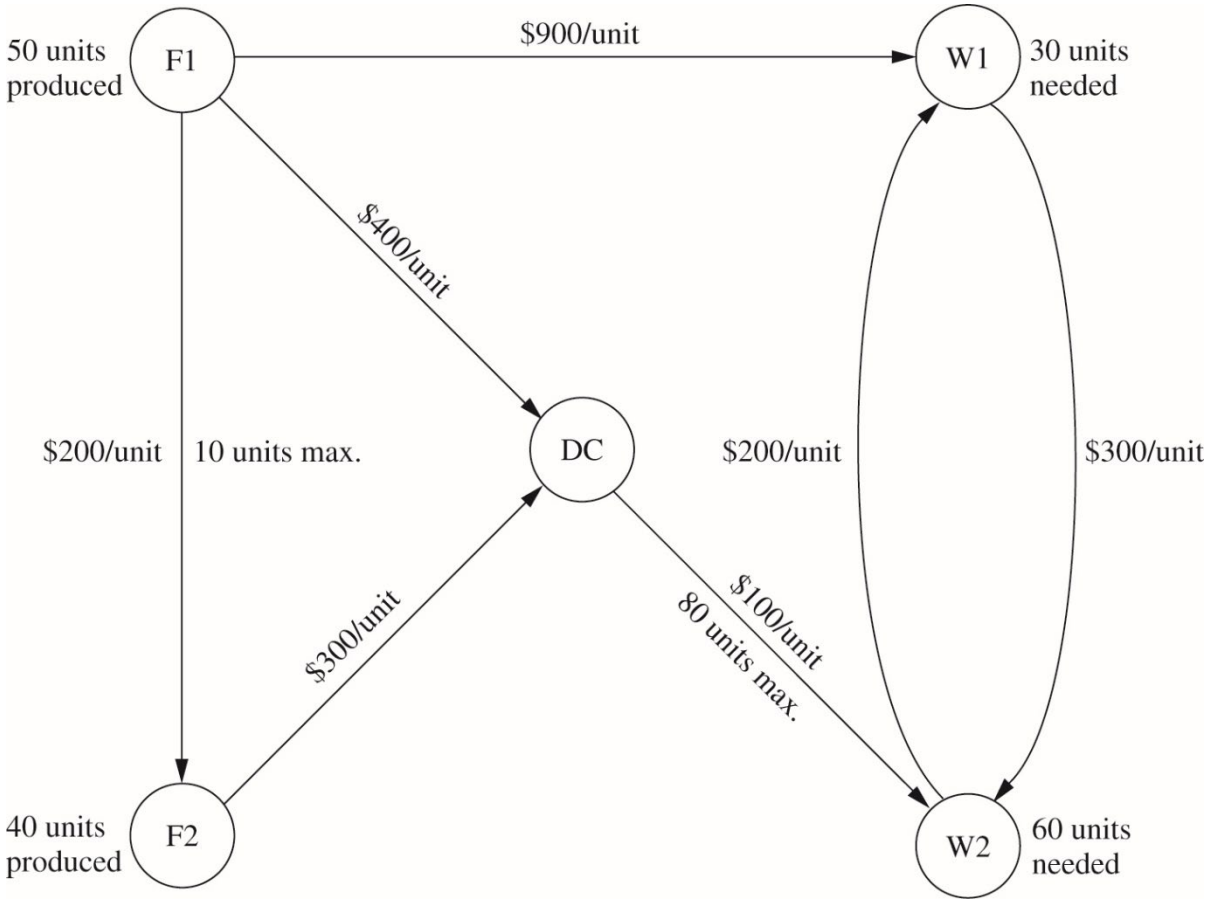
$$x_{ij} = \text{acres allocated to crop } i \in I \text{ at community } j \in J$$

- **Model**

$$\begin{array}{lll}
\max & \sum_{i\in I} R_i \sum_{j\in J} x_{ij} & \text{maximize total net return} \\
\text{s.t.} & \sum_{i\in I} x_{ij} \le L_j^U, \quad \forall j \in J, & \text{land constraints} \\
& \sum_{i\in I} W_i^C x_{ij} \le W_j^A, \quad \forall j \in J, & \text{water constraints} \\
& \sum_{j} x_{ij} \le L_i^Q, \quad \forall i \in I, & \text{land quota constraints} \\
& \dfrac{\sum_{i\in I} x_{ij}}{L_j^U} = z, \quad \forall j \in J, & \text{fairness} \\
& x_{ij} \ge 0, \quad \forall i \in I, j \in J. &
\end{array}$$

Example: Distribution Network

Distribution Network



Distribution Network

- Data

- Decisions

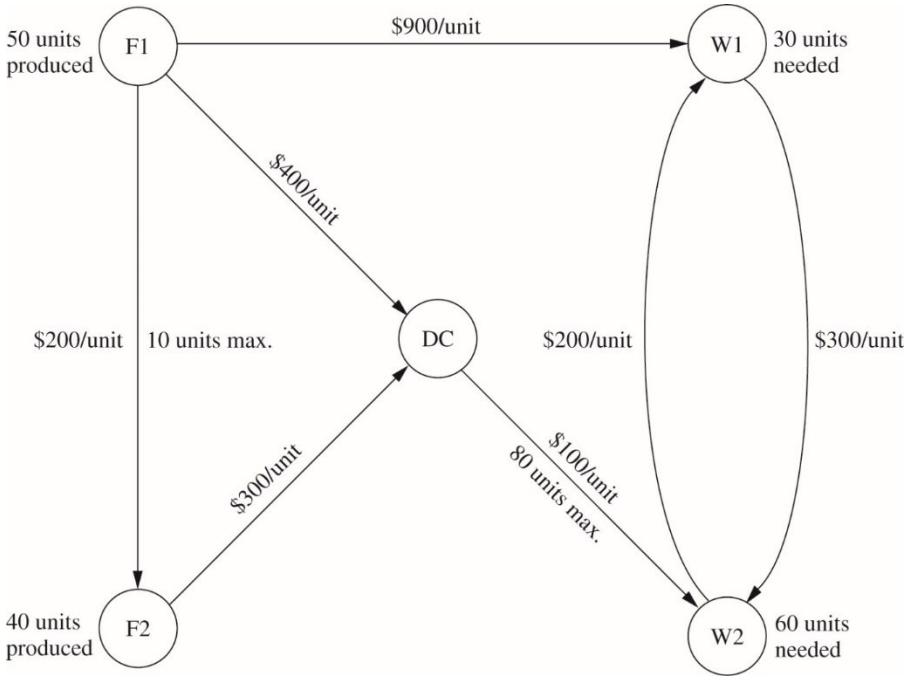
- Model

Distribution Network

- Data

- Decisions

- Model



*Distribution Network

- Data

$$\mathcal{G} = (\mathcal{N}, \mathcal{A}) = \text{the graph} \qquad \mathcal{A} = \text{set of arcs}$$

$$\mathcal{N} = \text{set of nodes} = \mathcal{N}^S \cup \mathcal{N}^D \cup \mathcal{N}^T$$

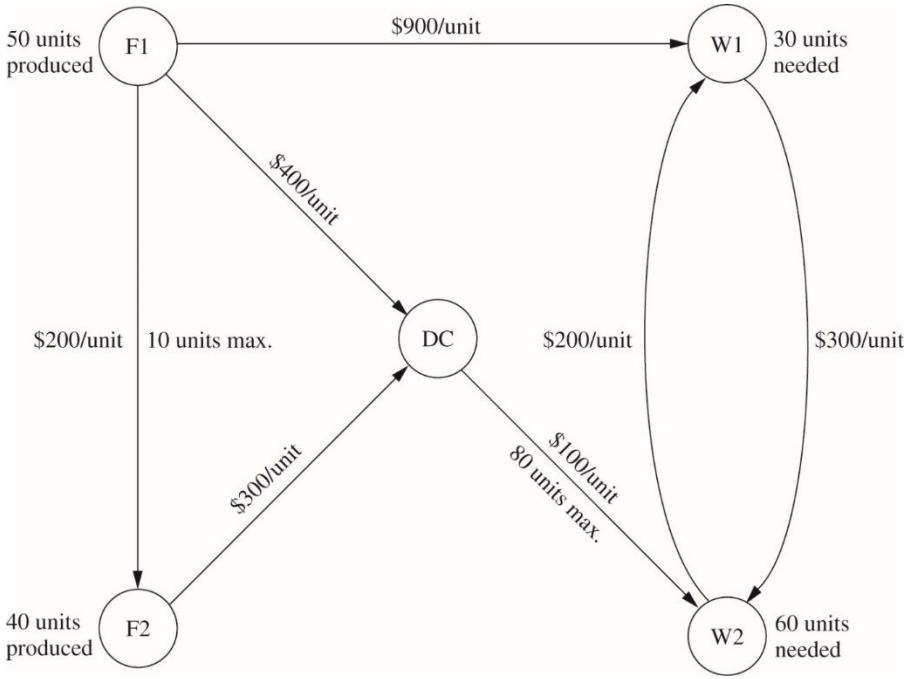
$$\mathcal{N}^S = \{F_1, F_2\}$$

$$\mathcal{N}^D = \{W_1, W_2\}$$

$$\mathcal{N}^T = \{\text{DC}\}$$

- Decisions

- Model



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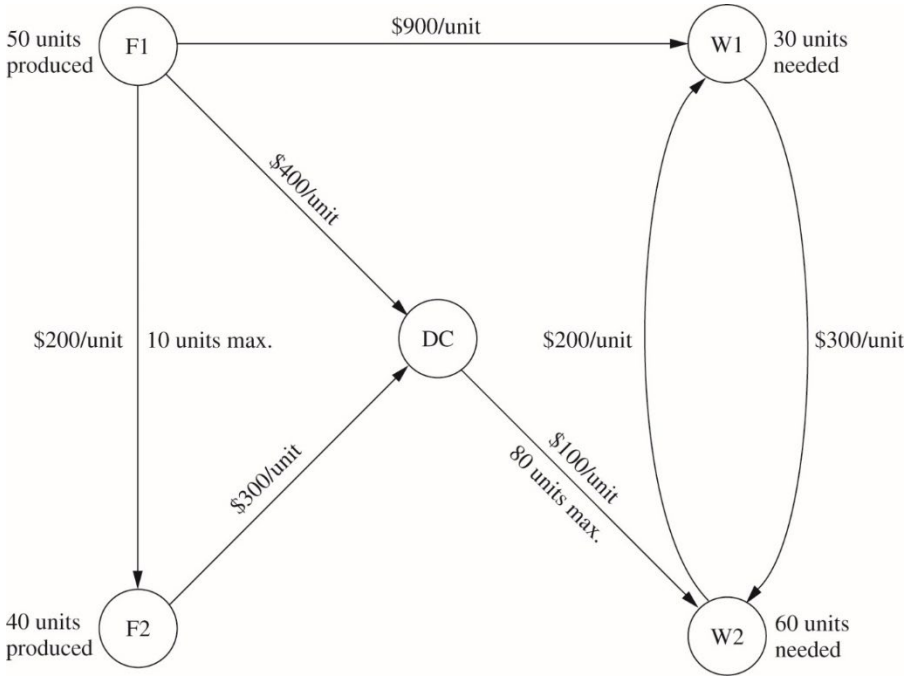
$$\mathcal{N}^T = \{\text{DC}\}$$

$$\mathcal{A} = \text{set of arcs}$$
$$c_{ij} = \text{unit cost of transportation on arc } (i,j) \in \mathcal{A}$$
$$Q_{ij} = \text{capacity on arc } (i,j) \in \mathcal{A}$$
$$S_i = \text{supply of products at node } i \in \mathcal{N}^S$$
$$D_j = \text{demand of products at node } j \in \mathcal{N}^D$$

- Decisions

- Model



*Distribution Network

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$$D_j = \text{demand of products at node } j \in \mathcal{N}^D$$

- Decisions

$$x_{ij} = \text{products transported on arc } (i,j)$$

- Model

*Distribution Network

Data

$\mathcal{G} = (\mathcal{N}, \mathcal{A})$ = the graph

$\mathcal{N}$ = set of nodes = $\mathcal{N}^S \cup \mathcal{N}^D \cup \mathcal{N}^T$

$\mathcal{N}^S = \{F_1, F_2\}$

$\mathcal{N}^D = \{W_1, W_2\}$

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$\mathcal{A}$ = set of arcs

c_{ij} = unit cost of transportation on arc $(i,j) \in \mathcal{A}$

Q_{ij} = capacity on arc $(i,j) \in \mathcal{A}$

S_i = supply of products at node $i \in \mathcal{N}^S$

D_j = demand of products at node $j \in \mathcal{N}^D$

Decisions

x_{ij} = products transported on arc (i,j)

Model

$$\min \sum_{(i,j)\in\mathcal{A}} c_{ij}x_{ij} \qquad \text{minimize total transportation costs}$$

s.t.

when the total supply equals the total demand

$$\left.\begin{aligned}
\sum_{\{i:(i,j)\in\mathcal{A}\}} x_{ij} - \sum_{\{k:(j,k)\in\mathcal{A}\}} x_{jk} &= D_j, && \forall j \in \mathcal{N}^D,\\
\sum_{\{i:(i,j)\in\mathcal{A}\}} x_{ij} - \sum_{\{k:(j,k)\in\mathcal{A}\}} x_{jk} &= 0, && \forall j \in \mathcal{N}^T,\\
\sum_{\{i:(i,j)\in\mathcal{A}\}} x_{ij} - \sum_{\{k:(j,k)\in\mathcal{A}\}} x_{jk} &= -S_j, && \forall j \in \mathcal{N}^S,
\end{aligned}\right\} \text{flow balance constraints}$$

$$0 \le x_{ij} \le Q_{ij}, \quad \forall (i,j) \in \mathcal{A}. \qquad \text{capacity constraints}$$

*Distribution Network

Data

$\mathcal{G} = (\mathcal{N}, \mathcal{A})$ = the graph

$\mathcal{N}$ = set of nodes = $\mathcal{N}^S \cup \mathcal{N}^D \cup \mathcal{N}^T$

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$\mathcal{A}$ = set of arcs

c_{ij} = unit cost of transportation on arc $(i,j) \in \mathcal{A}$

Q_{ij} = capacity on arc $(i,j) \in \mathcal{A}$

S_i = supply of products at node $i \in \mathcal{N}^S$

D_j = demand of products at node $j \in \mathcal{N}^D$

Decisions

x_{ij} = products transported on arc (i,j)

Model

$$\min \sum_{(i,j)\in\mathcal{A}} c_{ij}x_{ij} \qquad \text{minimize total transportation costs}$$

s.t.

when the total supply exceeds the total demand

$$\left.\begin{aligned}
\sum_{\{i:(i,j)\in\mathcal{A}\}} x_{ij} - \sum_{\{k:(j,k)\in\mathcal{A}\}} x_{jk} &= D_j, && \forall j \in \mathcal{N}^D,\\
\sum_{\{i:(i,j)\in\mathcal{A}\}} x_{ij} - \sum_{\{k:(j,k)\in\mathcal{A}\}} x_{jk} &= 0, && \forall j \in \mathcal{N}^T,\\
\sum_{\{i:(i,j)\in\mathcal{A}\}} x_{ij} - \sum_{\{k:(j,k)\in\mathcal{A}\}} x_{jk} &\geq -S_j, && \forall j \in \mathcal{N}^S,
\end{aligned}\right\} \text{flow balance constraints}$$

$$0 \leq x_{ij} \leq Q_{ij}, \quad \forall (i,j) \in \mathcal{A}. \qquad \text{capacity constraints}$$

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$\mathcal{G} = (\mathcal{N}, \mathcal{A})$ = the graph

$\mathcal{N}$ = set of nodes = $\mathcal{N}^S \cup \mathcal{N}^D \cup \mathcal{N}^T$

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D_j = demand of products at node $j \in \mathcal{N}^D$

Decisions

x_{ij} = products transported on arc (i,j)

Model

$$\min \sum_{(i,j)\in\mathcal{A}} c_{ij} x_{ij} \qquad \text{minimize total transportation costs}$$

s.t.

when the total demand exceeds the total supply

$$\left.\begin{aligned}
\sum_{\{i:(i,j)\in\mathcal{A}\}} x_{ij} - \sum_{\{k:(j,k)\in\mathcal{A}\}} x_{jk} &\le D_j, && \forall j \in \mathcal{N}^D, \\
\sum_{\{i:(i,j)\in\mathcal{A}\}} x_{ij} - \sum_{\{k:(j,k)\in\mathcal{A}\}} x_{jk} &= 0, && \forall j \in \mathcal{N}^T, \\
\sum_{\{i:(i,j)\in\mathcal{A}\}} x_{ij} - \sum_{\{k:(j,k)\in\mathcal{A}\}} x_{jk} &= -S_j, && \forall j \in \mathcal{N}^S,
\end{aligned}\right\} \text{flow balance constraints}$$

$$0 \le x_{ij} \le Q_{ij}, \quad \forall (i,j) \in \mathcal{A}. \qquad \text{capacity constraints}$$

Example: Production-Inventory Planning

Production-Inventory Planning

- Two types of vehicles: {sedan, wagon}
- Planning horizon: three months
- Demand
- Cost
  - Production cost
  - Inventory cost
- Initial inventory
- Constraints
  - Production capacity: 1,500 vehicles / month
  - Month 1: at least 2/3 of cars produced are sedan.

	Market demand (vehicle)	
Month	Sedan	Wagon
1	1100	600
2	1500	700
3	1200	500

	Sedan	Wagon
Price (/vehicle)	$8,000	$9,000
Production cost (/vehicle)	$6,000	$7,500
Inventory cost (/vehicle • month)	$150	$200
Initial inventory (vehicle)	200	100

- Decision: Production plan
- Objective: Maximize profit

Production-Inventory Planning

- Data
- Decisions
- Model

Production-Inventory Planning

- Data

- Decisions

- Model

	Market demand (vehicle)	
Month	Sedan	Wagon
1	1100	600
2	1500	700
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	Sedan	Wagon
Price (/vehicle)	$8,000	$9,000
Production cost (/vehicle)	$6,000	$7,500
Inventory cost (/vehicle • month)	$150	$200
Initial inventory (vehicle)	200	100

*Production-Inventory Planning

Data

p_i = price of vehicle type $i \in I = \{\text{sedan}, \text{wagon}\}$

c_i = production cost of vehicle type $i \in I$

h_i = inventory cost of vehicle type $i \in I$

α = minimum ratio of sedan production in month 1

Q = production capacity

I_{0i} = initial inventory of vehicle type $i \in I$

D_{ti} = demand of vehicle type $i \in I$ in month $t \in T = \{1, 2, 3\}$

Decisions

Model

	Market demand (vehicle)	
Month	Sedan	Wagon
1	1100	600
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	Sedan	Wagon
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- Data

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- Decisions

x_{ti} = number of vehicle type $i \in I$ produced in month $t \in T$

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*Production-Inventory Planning

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- Decisions

x_{ti} = number of vehicle type $i \in I$ produced in month $t \in T$

s_{ti} = sales of vehicle type $i \in I$ in month $t \in T$

I_{ti} = inventory of vehicle type $i \in I$ at the end of month $t \in T$

- Model

*Production-Inventory Planning

TOpS

Data

p_i = price of vehicle type $i \in I = \{\text{sedan}, \text{wagon}\}$

c_i = production cost of vehicle type $i \in I$

h_i = inventory cost of vehicle type $i \in I$

α = minimum ratio of sedan production in month 1

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I_{0i} = initial inventory of vehicle type $i \in I$

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Decisions

x_{ti} = number of vehicle type $i \in I$ produced in month $t \in T$

s_{ti} = sales of vehicle type $i \in I$ in month $t \in T$

I_{ti} = inventory of vehicle type $i \in I$ at the end of month $t \in T$

Model

$$
\begin{array}{lrll}
\max & \sum_{i\in I} p_i \sum_{t\in T} s_{ti} - \sum_{i\in I} c_i \sum_{t\in T} x_{ti} - \sum_{i\in I} h_i \sum_{t\in T} I_{ti} & & \text{revenue} - \text{production cost} - \text{inventory cost} \\
\text{s.t.} & \sum_{i\in I} x_{ti} \le Q, & \forall t \in T, & \text{production capacity constraints} \\
& s_{ti} \le I_{t-1,i} + x_{ti}, & \forall t \in T, i \in I, & \text{product availability constraints} \\
& s_{ti} \le D_{ti}, & \forall t \in T, i \in I, & \text{demand constraints} \\
& I_{t-1,i} + x_{ti} - s_{ti} = I_{ti}, & \forall t \in T, i \in I, & \text{inventory balance constraints} \\
& (1-\alpha)x_{11} - \alpha x_{12} \ge 0, & & \text{production restriction of month 1} \\
& s_{ti}, I_{ti}, x_{ti} \ge 0, & \forall t \in T, i \in I. &
\end{array}
$$

*Production-Inventory Planning

Data

p_i = price of vehicle type $i \in I = \{\text{sedan}, \text{wagon}\}$

c_i = production cost of vehicle type $i \in I$

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Model

$$
\begin{array}{lll}
\max & \sum_{i \in I} p_i \sum_{t \in T} s_{ti} - \sum_{i \in I} c_i \sum_{t \in T} x_{ti} - \sum_{i \in I} h_i \sum_{t \in T} I_{ti} & \text{revenue} - \text{production cost} - \text{inventory cost} \\
\text{s.t.} & \sum_{i \in I} x_{ti} \le Q, \quad \forall t \in T, & \text{production capacity constraints} \\
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Model

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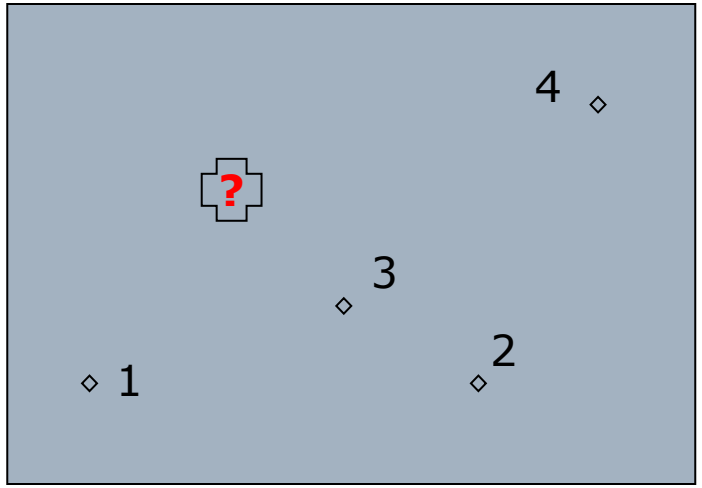
\begin{array}{lrll}
\max & \sum_{i \in I} p_i \sum_{t \in T} s_{ti} - \sum_{i \in I} c_i \sum_{t \in T} x_{ti} - \sum_{i \in I} h_i \sum_{t \in T} I_{ti} & & \text{revenue} - \text{production cost} - \text{inventory cost} \\
\text{s.t.} & \sum_{i \in I} x_{ti} \le Q, & \forall t \in T, & \text{production capacity constraints} \\
& s_{ti} \le D_{ti}, & \forall t \in T, i \in I, & \text{demand constraints} \\
& I_{t-1,i} + x_{ti} - s_{ti} = I_{ti}, & \forall t \in T, i \in I, & \text{inventory balance constraints} \\
& (1-\alpha)x_{11} - \alpha x_{12} \ge 0, & & \text{production restriction of month 1} \\
& s_{ti}, I_{ti}, x_{ti} \ge 0, & \forall t \in T, i \in I. &
\end{array}
$$

Example: Facility Location

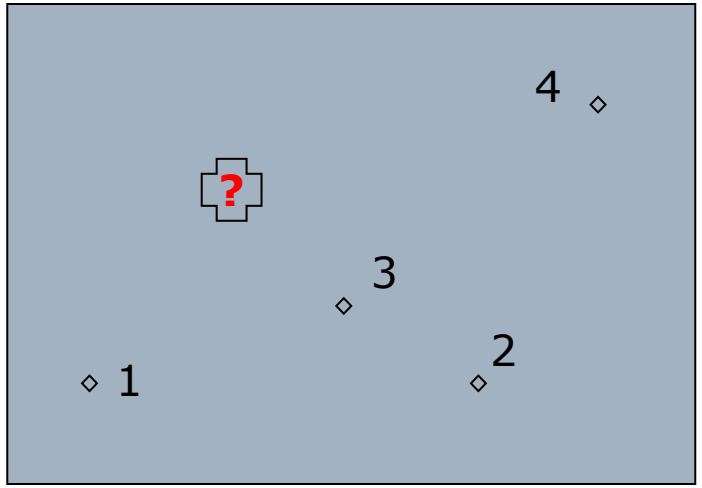
Facility Location

- Four towns
  - Coordinates
    1(10, 20), 2(60,20), 3(40,30), 4(80,60)
  - Fire frequencies
    1(20), 2(30), 3(40), 4(25)
- Decision
  - Locate one fire station
- Constraint
  - Travel only east-west or north-south
- Objective
  - Minimize the total (weighted) travel distance



Facility Location

- Data
- Decisions
- Model



*Facility Location

- Data

$$(x_i, y_i) = \text{the position of town } i \in I = \{1, 2, 3, 4\}$$
$$f_i = \text{fire alarm frequency of town } i \in I$$

- Decisions

$$(x, y) = \text{the position of fire station}$$

- Model

*Facility Location

- Data

$$(x_i, y_i) = \text{the position of town } i \in I = \{1, 2, 3, 4\}$$
$$f_i = \text{fire alarm frequency of town } i \in I$$

- Decisions

$$(x, y) = \text{the position of fire station}$$

- Model

$$\boxed{\min \quad \sum_{i \in I} f_i(|x - x_i| + |y - y_i|)}$$

*Facility Location

- Data

$$(x_i, y_i) = \text{the position of town } i \in I = \{1, 2, 3, 4\}$$
$$f_i = \text{fire alarm frequency of town } i \in I$$

- Decisions

$$(x, y) = \text{the position of fire station}$$

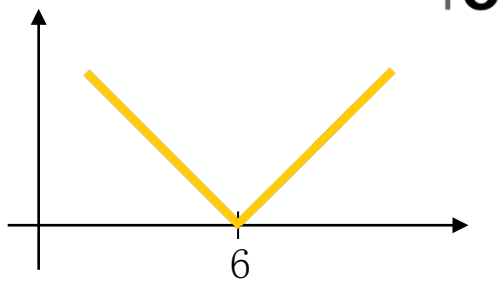
- Model

$$\boxed{\min \quad \sum_{i \in I} f_i(|x - x_i| + |y - y_i|)}$$

Nonlinear!

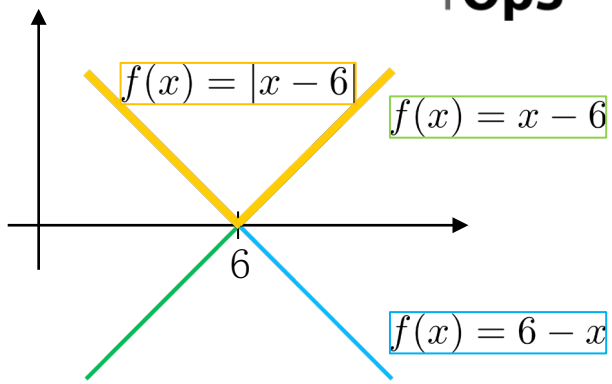
*Facility Location

$$\min \quad f(x) = |x - 6|$$



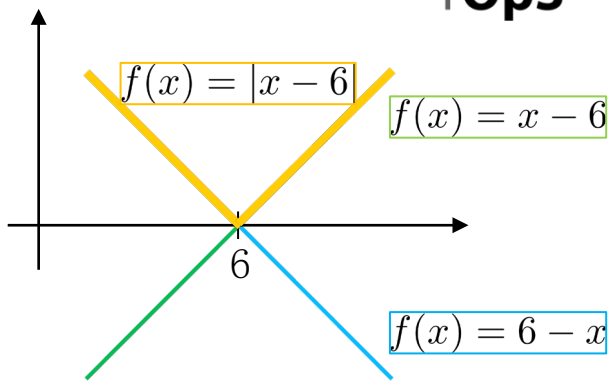
*Facility Location

$$\min \quad f(x) = |x-6|$$



*Facility Location

$$\min \quad f(x) = |x-6|$$



- Model 1

$$\begin{aligned} \min \quad & z \\ \text{s.t.} \quad & x-6 \le z, \\ & 6-x \le z. \end{aligned}$$

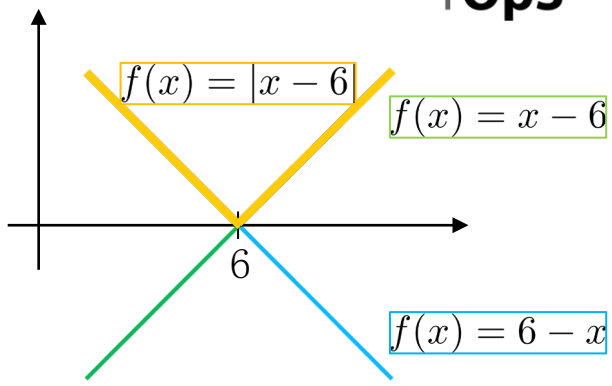
*Facility Location

- Data

$$(x_i, y_i) = \text{the position of town } i \in I = \{1, 2, 3, 4\}$$
$$f_i = \text{fire alarm frequency of town } i \in I$$

- Decisions

$$(x, y) = \text{the position of fire station}$$



- Model

- Model 1

$$\begin{aligned} \min \quad & z \\ \text{s.t.} \quad & x - 6 \le z, \\ & 6 - x \le z. \end{aligned}$$

*Facility Location

- Data

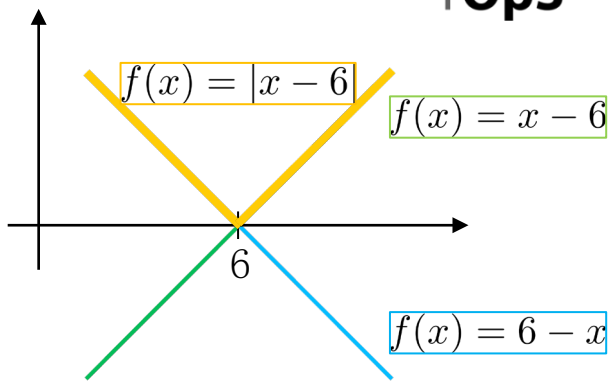
$$(x_i, y_i) = \text{the position of town } i \in I = \{1, 2, 3, 4\}$$
$$f_i = \text{fire alarm frequency of town } i \in I$$

- Decisions

$$(x, y) = \text{the position of fire station}$$

- Model

$$\boxed{\begin{array}{lll} \min & \sum_{i \in I} f_i(x'_i + y'_i) & \\ \text{s.t.} & x - x_i \le x'_i, & \forall i \in I, \\ & x_i - x \le x'_i, & \forall i \in I, \\ & y - y_i \le y'_i, & \forall i \in I, \\ & y_i - y \le y'_i, & \forall i \in I. \end{array}}$$



- Model 1

$$\begin{array}{ll} \min & z \\ \text{s.t.} & x - 6 \le z, \\ & 6 - x \le z. \end{array}$$

*Facility Location

$$\min \quad f(x) = |x - 6|$$

*Facility Location

$$\min \quad f(x) = |x - 6|$$

- Model 2

$$\begin{aligned} \min \quad & (x^+ + x^-) \\ \text{s.t.} \quad & x - 6 = x^+ - x^-, \\ & x^+, x^- \geq 0. \end{aligned}$$

*Facility Location

- Data

$$(x_i, y_i) = \text{the position of town } i \in I = \{1, 2, 3, 4\}$$
$$f_i = \text{fire alarm frequency of town } i \in I$$

- Decisions

$$(x, y) = \text{the position of fire station}$$

- Model

- Model 2

$$\begin{aligned} \min \quad & (x^+ + x^-) \\ \text{s.t.} \quad & x - 6 = x^+ - x^-, \\ & x^+, x^- \geq 0. \end{aligned}$$

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$$f_i = \text{fire alarm frequency of town } i \in I$$

- Decisions

$$(x, y) = \text{the position of fire station}$$

- Model

$$\boxed{\begin{aligned} \min \quad & \sum_{i \in I} f_i \left[(x_i^+ + x_i^-) + (y_i^+ + y_i^-) \right] \\ \text{s.t.} \quad & x - x_i = x_i^+ - x_i^-, \quad \forall i \in I, \\ & y - y_i = y_i^+ - y_i^-, \quad \forall i \in I, \\ & x_i^+, x_i^-, y_i^+, y_i^- \geq 0, \quad \forall i \in I. \end{aligned}}$$

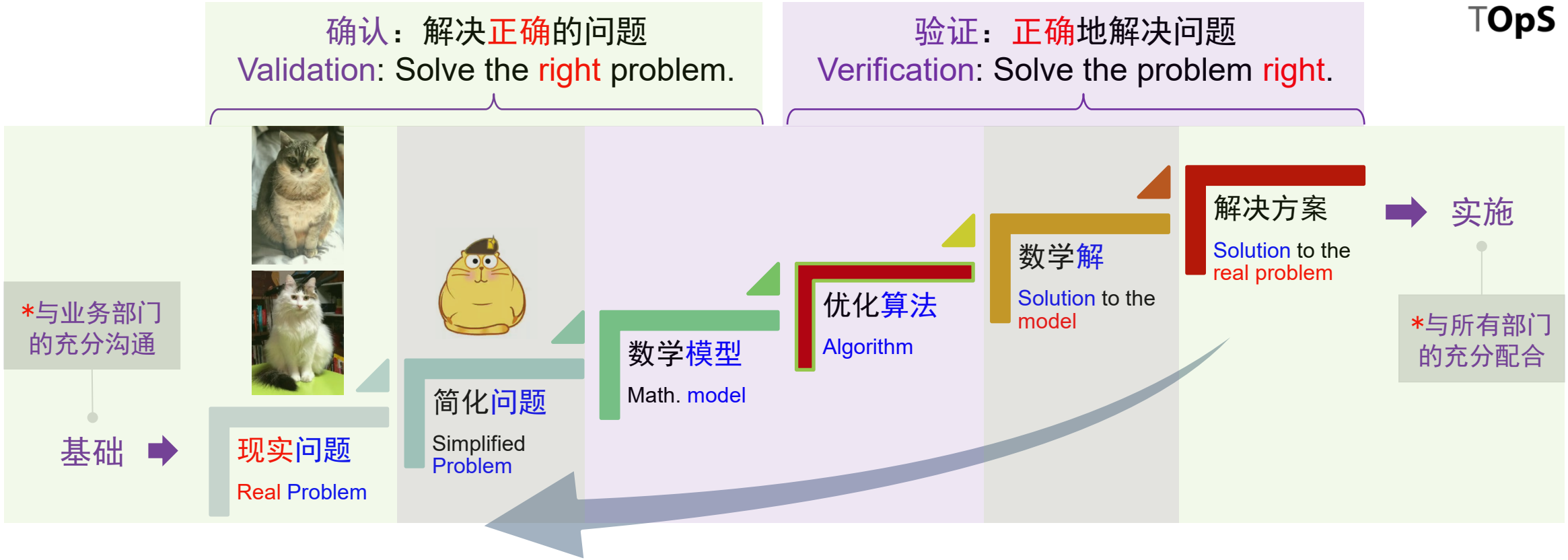
- Model 2

$$\begin{aligned} \min \quad & (x^+ + x^-) \\ \text{s.t.} \quad & x - 6 = x^+ - x^-, \\ & x^+, x^- \geq 0. \end{aligned}$$

Linearization Techniques: Important

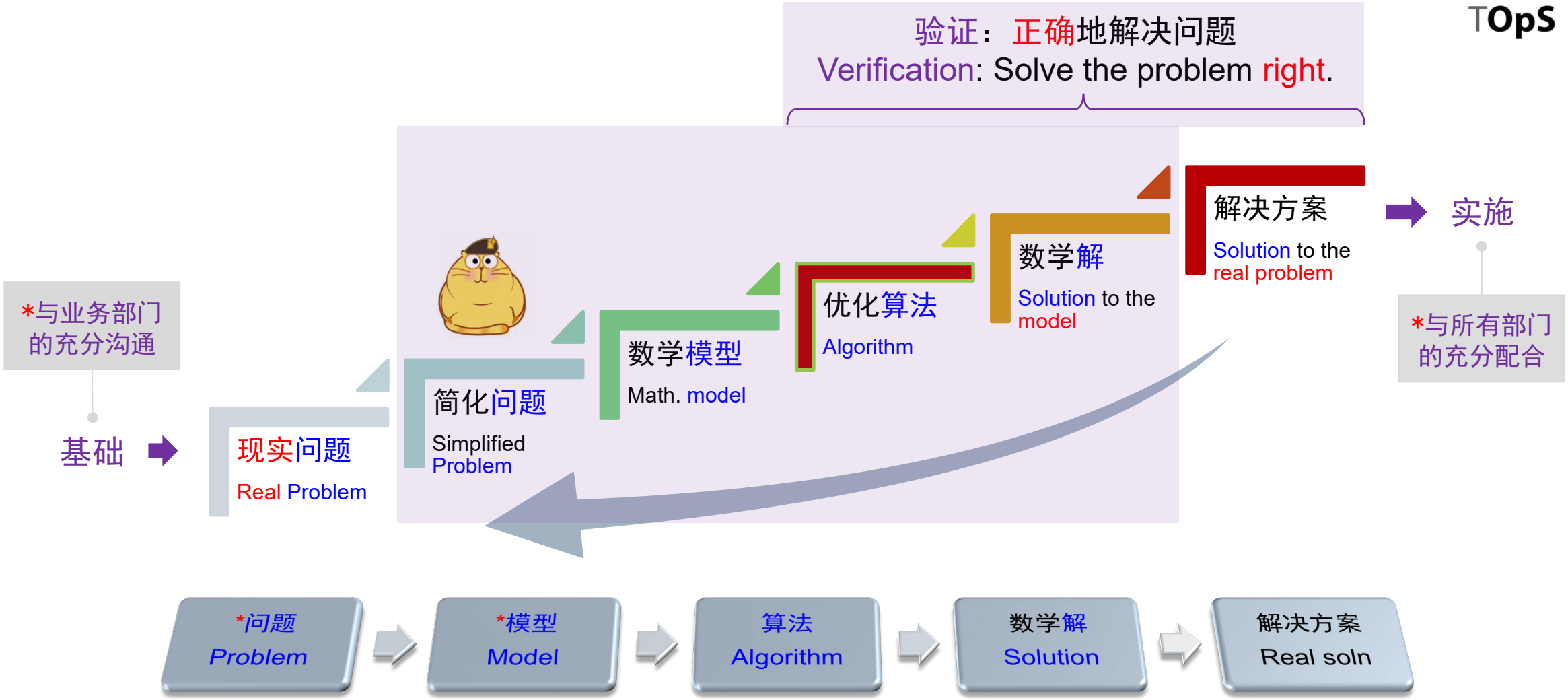
*Recap

*OR Application Roadmap



Source: TOpS Lab (2020)

*OR Application Roadmap – This Course



Source: TOpS Lab (2020)

*Optimization Problem/Model: Key Elements

- 我们知道什么? What do we know?

  流程 + 数据 process + data

- 我们需要决定什么？What do we need to decide?

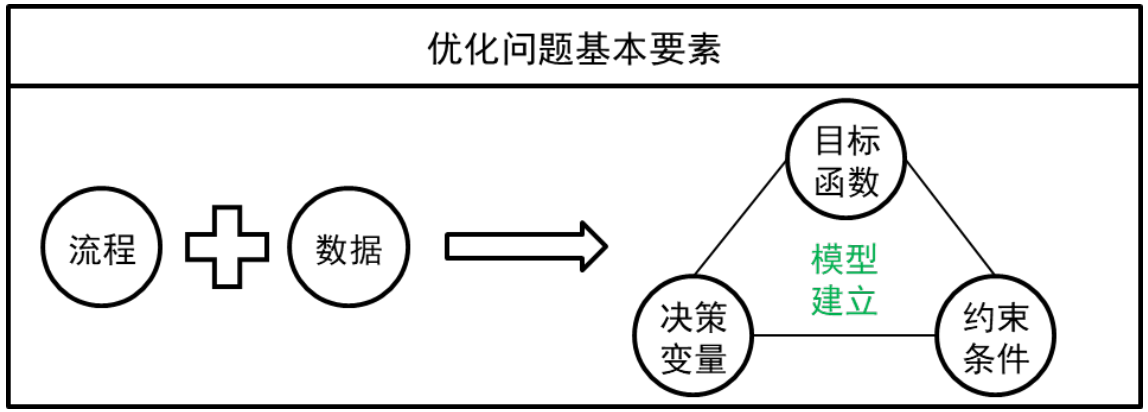
  决策变量 decision variables

- 我们希望达到什么目标？What do we want to achieve?

  目标函数 objective function

- 我们的决策受到什么限制？What prohibits our ability to do so?

  约束条件 constraints



Resource Allocation Problems

- Allocating limited resources among competing activities in a best possible (i.e. *optimal*) way.

- Activities ➔ levels of activities ➔ consumption of resources
  - Decision variables
- Limited resources ➔ necessary to perform activities
  - Constraints
- Overall measure of performance
  - Objective function

Source: Hillier and Lieberman (2021)

Wyndor Glass Co.

- Two products
  - Product 1
    - Glass door w/ aluminum framing
  - Product 2
    - Wood-frame window
- Three plants
  - Plant 1
    - Aluminum frames and hardware
  - Plant 2
    - Wood frames
  - Plant 3
    - Glass and product assemble
- Data
  - Available production time in each plant (hours / week)
  - Production time used in each plant for each batch of each product (hours / batch)
  - Profit per batch of each product

	Doors	Windows		
Profit per batch	$3,000	$5,000		
				Hours
	Hours used per batch produced			available
Plant 1	1	0		4
Plant 2	0	2		12
Plant 3	3	2		18

- What do we need to decide?
- How do we evaluate our decision?

*Wyndor Glass Co.

- Mathematical formulation
  - x_1 = # batches of product 1 produced per week
  - x_2 = # batches of product 2 produced per week
  - Z = total profit per week

	Doors	Windows			
Profit per batch	$3,000	$5,000			
					Hours
	Hours used per batch produced				available
Plant 1	1	0			4
Plant 2	0	2			12
Plant 3	3	2			18
	Doors	Windows			Total profit
Batches produced					

$$
\begin{aligned}
\max \quad & Z = 3x_1 + 5x_2 \\
\text{s.t.} \quad & x_1 \le 4, \\
& 2x_2 \le 12, \\
& 3x_1 + 2x_2 \le 18, \\
& x_1, \quad x_2 \ge 0.
\end{aligned}
$$

Lingo Model - 4.1_wyndor_lingo

```
! The Wyndor Model

! x1 = doors, x2 = windows.
! objective;
max = 3*x1 + 5*x2;

! constraints @ plants 1, 2, & 3;
        x1          < 4;
            2*x2    < 12;
      3*x1 + 2*x2   < 18;
end
```

<=

Standard Form of LP Models

Maximize $Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n$

Subject to

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &\le b_1, \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &\le b_2, \\
&\vdots \\
a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &\le b_m,
\end{aligned}
$$

and

$$x_1 \ge 0, \quad x_2 \ge 0, \quad \cdots, \quad x_n \ge 0.$$

Maximize $Z = c^{\mathrm{T}}x$

Subject to

$$
\begin{aligned}
Ax &\le b, \\
x &\ge 0.
\end{aligned}
$$

Standard Form of LP Models

Maximize $Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n$

Subject to

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \le b_1,$$
$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \le b_2,$$
$$\vdots$$
$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \le b_m,$$

≥ 0

and

$$x_1 \ge 0, \quad x_2 \ge 0, \quad \cdots, \quad x_n \ge 0.$$

Maximize $Z = c^{\mathrm{T}}x$

Subject to

$$Ax \le b,$$
$$x \ge 0.$$

≥ 0

Four Assumptions of LP

- Proportionality
  - Objective value/LHS is proportional to the level of activity
  - Nonlinear/mixed integer programming
- Additivity
  - Sum of individual contributions (objective/LHS)
  - Nonlinear programming
- Divisibility
  - (Mixed) Integer programming
- Certainty
  - Sensitivity analysis, but …
  - Stochastic programming, (approximate) dynamic programming, simulation optimization, robust optimization, …

In a linear programming (LP) model, the objective function and (the left-hand-side (LHS) functions of) the constraints must be linear functions of the decision variables.

Mathematical Programming Solvers

- Excel
- Lindo/Lingo
- IBM ILOG Cplex
- FICO Xpress
- **Gurobi**
- SAS
- AIMMS
- GAMS
- Frontline

$$
\begin{array}{lrl}
\max & Z = 3x_1 + 5x_2 & \\
\text{s.t.} & x_1 & \leq 4, \\
 & 2x_2 & \leq 12, \\
 & 3x_1 + 2x_2 & \leq 18, \\
 & x_1, \quad x_2 & \geq 0.
\end{array}
$$

- Free/open source solvers

	Doors	Windows			
Profit per batch	$3,000	$5,000			
			Hours		Hours
	Hours used per batch produced		used		available
Plant 1	1	0	2	<=	4
Plant 2	0	2	12	<=	12
Plant 3	3	2	18	<=	18
	Doors	Windows			Total profit
Batches produced	2	6			$36,000

Lingo Model - 4.1_wyndor_lingo

```
! The Wyndor Model

! x1 = doors, x2 = windows.
! objective;
max = 3*x1 + 5*x2;

! constraints @ plants 1, 2, & 3;
        x1          < 4;
             2*x2 < 12;
      3*x1 + 2*x2 < 18;
end
```

wyndor.py

```
from wyndor_pandas import generate_data
from wyndor_gurobi import formulate_model, solve_model, write_model
#from wyndor_gurobi import formulate_model, read_model, solve_model, write_model

# Generate Pandas DataFrames directly.
products, products_plants, plants = generate_data()

# Formulate the LP instance
# Option 1: formulate manually
model = formulate_model(products, products_plants, plants)
# Option 2: read from a .lp or .mps file
# model = read_model("wyndor_read.lp")
# model = read_model("wyndor_read.mps")

# Write the LP instance as a .lp or .mps file.
write_model(model, "wyndor_write.lp")
# write_model(model, "wyndor_write.mps")

# Solve the LP instance.
solve_model(model)
```

Source: http://www.lindo.com/

Modeling Examples

Regional planning

$$
\begin{array}{lll}
\max & \sum_{i\in I} R_i \sum_{j\in J} x_{ij} & \text{maximize total net return} \\
\text{s.t.} & \sum_{i\in I} x_{ij} \le L_j^U, \quad \forall j \in J, & \text{land constraints} \\
& \sum_{i\in I} W_i^C x_{ij} \le W_j^A, \quad \forall j \in J, & \text{water constraints} \\
& \sum_{j} x_{ij} \le L_i^Q, \quad \forall i \in I, & \text{land quota constraints} \\
& \frac{\sum_{i\in I} x_{ij}}{L_j^U} = z, \quad \forall j \in J, & \text{fairness} \\
& x_{ij} \ge 0, \quad \forall i \in I, j \in J. &
\end{array}
$$

Production-inventory planning

$$
\begin{array}{lll}
\max & \sum_{i\in I} p_i \sum_{t\in T} s_{ti} - \sum_{i\in I} c_i \sum_{t\in T} x_{ti} - \sum_{i\in I} h_i \sum_{t\in T} I_{ti} & \text{revenue} - \text{production cost} - \text{inventory cost} \\
\text{s.t.} & \sum_{i\in I} x_{ti} \le Q, \quad \forall t \in T, & \text{production capacity constraints} \\
& s_{ti} \le D_{ti}, \quad \forall t \in T, i \in I, & \text{demand constraints} \\
& I_{t-1,i} + x_{ti} - s_{ti} = I_{ti}, \quad \forall t \in T, i \in I, & \text{inventory balance constraints} \\
& (1-\alpha)x_{11} - \alpha x_{12} \ge 0, & \text{production restriction of month 1} \\
& s_{ti}, I_{ti}, x_{ti} \ge 0, \quad \forall t \in T, i \in I. &
\end{array}
$$

Distribution network

$$
\begin{array}{lll}
\min & \sum_{(i,j)\in\mathcal{A}} c_{ij} x_{ij} & \text{minimize total transportation costs} \\
\text{s.t.} & & \\
& \left.\begin{array}{ll}
\sum_{\{i:(i,j)\in\mathcal{A}\}} x_{ij} - \sum_{\{k:(j,k)\in\mathcal{A}\}} x_{jk} = D_j, & \forall j \in \mathcal{N}^D, \\
\sum_{\{i:(i,j)\in\mathcal{A}\}} x_{ij} - \sum_{\{k:(j,k)\in\mathcal{A}\}} x_{jk} = 0, & \forall j \in \mathcal{N}^T, \\
\sum_{\{i:(i,j)\in\mathcal{A}\}} x_{ij} - \sum_{\{k:(j,k)\in\mathcal{A}\}} x_{jk} \ge -S_j, & \forall j \in \mathcal{N}^S,
\end{array}\right\} & \text{flow balance constraints} \\
& 0 \le x_{ij} \le Q_{ij}, \quad \forall (i,j) \in \mathcal{A}. & \text{capacity constraints}
\end{array}
$$

Facility location

$$
\begin{array}{ll}
\min & \sum_{i\in I} f_i(x_i' + y_i') \\
\text{s.t.} & x - x_i \le x_i', \quad \forall i \in I, \\
& x_i - x \le x_i', \quad \forall i \in I, \\
& y - y_i \le y_i', \quad \forall i \in I, \\
& y_i - y \le y_i', \quad \forall i \in I.
\end{array}
$$

$$
\begin{array}{ll}
\min & \sum_{i\in I} f_i \left[(x_i^+ + x_i^-) + (y_i^+ + y_i^-)\right] \\
\text{s.t.} & x - x_i = x_i^+ - x_i^-, \quad \forall i \in I, \\
& y - y_i = y_i^+ - y_i^-, \quad \forall i \in I, \\
& x_i^+, x_i^-, y_i^+, y_i^- \ge 0, \quad \forall i \in I.
\end{array}
$$

Q & A